

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 4 Maths — CA-1 (Hard)

Class : 4

Subject : Maths (Math-Mela)

Max Marks : 30

Duration : 1 Hour

Syllabus: Chapters 1–3 — Shapes Around Us, Hide and Seek, Pattern Around Us

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

SECTION A — Multiple Choice Questions (1 Mark each)

- A solid shape has 5 faces, 9 edges, and 6 corners. Using Faces + Corners - Edges = 2, verify if this could be a valid solid, and identify it. (1 Mark)
A. $5 + 6 - 9 = 2$, valid; likely a triangular prism
B. $5 + 6 - 9 = 0$, invalid
C. $5 + 9 - 6 = 8$, invalid
D. The formula cannot be used here
- If you see only the top view and front view of an object but not the side view, what limitation might you face in fully understanding its shape? (1 Mark)
A. No limitation, two views are always enough
B. You might not know the exact depth or shape variations only visible from the side
C. Top and front views are identical, so no information is lost
D. It becomes impossible to know anything about the object
- A sequence of numbers goes 2, 4, 6, 8, What is the 10th number in this pattern? (1 Mark)
A. 18
B. 20
C. 22
D. 16
- Which of these combinations of coins/notes to make ₹8377;145 uses the FEWEST number of pieces (available: 100, 50, 20, 10, 5)? (1 Mark)
A. $100 + 20 + 20 + 5$
B. $50 + 50 + 20 + 20 + 5$
C. $100 + 50 - 5$ (not valid)
D. $20 \times 7 + 5$
- A pyramid has a square base. How many triangular faces will it have in addition to its base? (1 Mark)
A. 2
B. 3
C. 4
D. 5
- If a grid position is described as "2 rows down and 3 columns right" from the top-left corner, which of these best describes this position? (1 Mark)
A. Row 2, Column 3
B. Row 3, Column 2
C. Row 1, Column 1
D. It cannot be determined
- Which statement is TRUE about the sum of two odd numbers? (1 Mark)
A. The sum is always odd
B. The sum is always even
C. The sum can be either odd or even
D. Odd numbers cannot be added together

SECTION B — Short Answer Questions (2 Marks each)

- Explain why knowing only the top view of a building is not enough to fully construct it, using reasoning about missing information. (0 Marks)
- A pattern of numbers goes 3, 7, 11, 15, Identify the rule of the pattern and find the next two numbers. (0 Marks)
- Explain why a triangular pyramid (with a triangular base) has fewer faces than a square pyramid (with a square base), using their properties. (0 Marks)
- A group of 47 students is being divided for two different games - one needing pairs, and one needing groups of 5. Explain which grouping will have students left over and why. (0 Marks)
- Explain how understanding both grid positions and directions (like up, down, left, right) together helps in describing movement, such as in a simple maze. (0 Marks)
- A cube net (unfolded cube) is shown with 6 squares arranged in a cross shape. Explain how you would verify that this net can actually fold into a cube. (0 Marks)

SECTION C — Short Answer Questions (Explain) (3 Marks each)

- Analyse how the properties of 3D shapes (faces, edges, corners) follow a predictable mathematical relationship (Faces + Corners - Edges = 2), and discuss why this rule is useful for verifying whether a described shape is possible. (0 Marks)
- Discuss how combining knowledge of views (top, front, side) with grid-based positioning creates a more complete spatial understanding than either skill alone. (0 Marks)

SECTION D — Long Answer Questions (5 Marks each)

1. Write a detailed essay analysing how the three chapters studied (Shapes Around Us, Hide and Seek, Pattern Around Us) connect to real-world careers or fields, such as architecture, design, or data analysis, explaining the specific skills from each chapter that would be useful. (0 Marks)